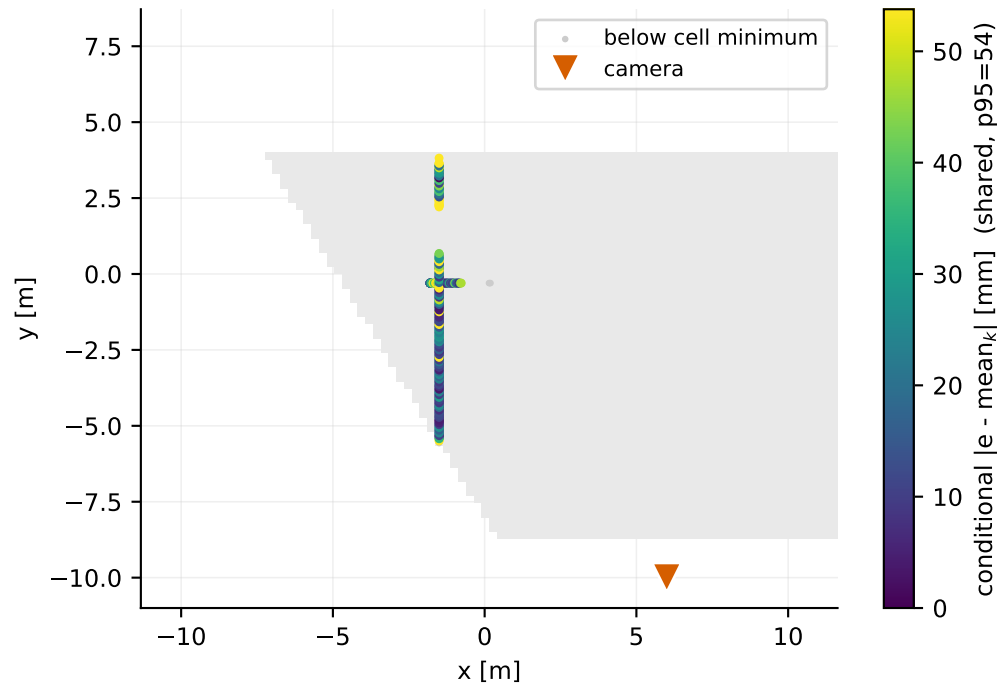
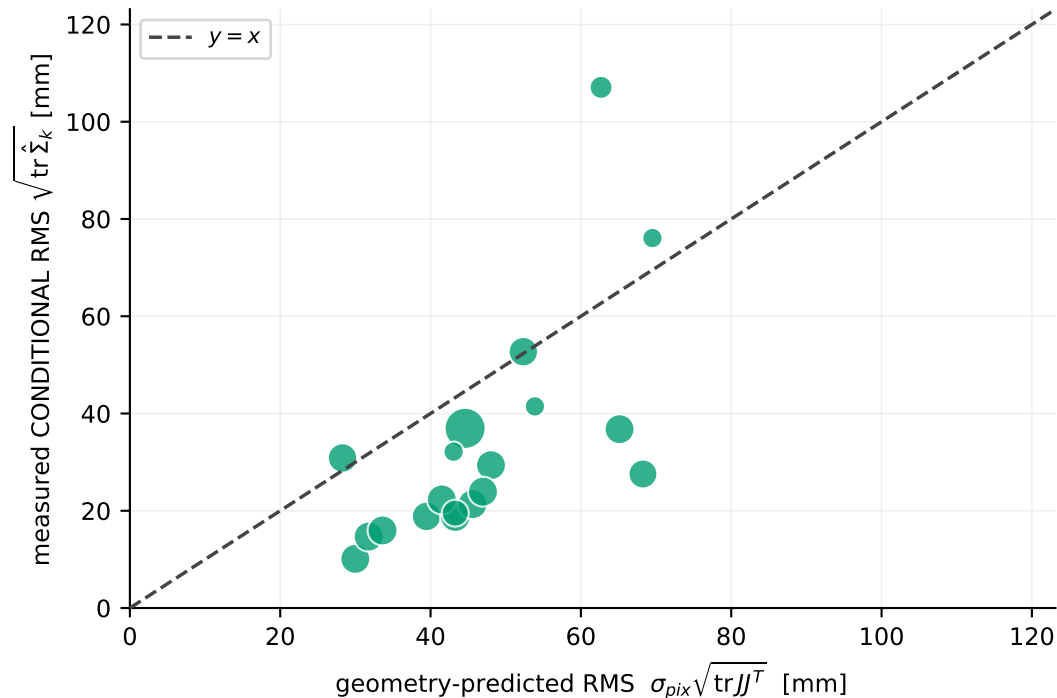


Camera C — deployed-corrected residual

(a) where the robot was actually observed
colour = error AFTER removing the local mean (bias split out)



(b) does geometry predict the local SPREAD?
N = 474 detections · M = 19 cells (0.5 m, ≥ 5 each)
conditional: R^2 vs identity = +0.07 · cell Spearman = +0.70 · median obs/pred = 0.54
median per-cell BIAS removed first = 85 mm



Point area in (b) scales with detections in the cell. σ_{pix} is fitted on leave-region-out folds, so each cell's prediction is out-of-region. Ground truth positions the dots in (a) and measures the residual; it never enters a projection, Jacobian or covariance. **READ THE LIMIT BEFORE THE RESULT:** the sampled ribbon in (a) never covers the footprint in two dimensions, so range, image row and projection conditioning are collinear along it. A weak or negative R^2 here therefore does NOT show that geometry fails -- it shows this dataset cannot test it. The honest conclusion is that straight-route logs are insufficient to decide whether full 2-D projection geometry beats a simpler range effect; that needs a commissioning capture with 2-D coverage and multiple headings per location.